

Huge library of components. Altium Designer delivers over 95,000 components out of the box. Not only are all the parts IPC-compliant, but the library also includes a table-driven IPC footprint builder. Altium Designer has live links with Farnell, Digi-Key and Newark to automatically update and populate your library with component information.

To sum up

Kramer says, "The integration of electronic and mechanical design is inexorable; with a few exceptions, PCB design is influenced not only by the components it carries but also by the space it can occupy. Many products today feature a single PCB and in such cases the size and shape of the PCB are dictated less by its functionality and more by its environment. In fact, in some cases—particularly in consumer devices—the shape and size of the end product will actually define the available space for the PCB and all its components.

E-CAD meets M-CAD

"While the focus of electronic CAD tool vendors has, understandably, been on addressing the complexity of electronic design, their counterparts have been industrious in improving M-CAD tools to make full use of the processing and graphical capabilities of the latest PCs and desktop computers. It is now commonplace for mechanical design engineers to have access to three-dimensional representations of their designs, rendered in real-time. As a productivity enhancement it is hard to deny the value of seeing the product of an engineer's efforts in 3D environment that support real-time manipulation to vary the viewing angle.

It is also valid to mention that while ICs continue to shrink in size, their supporting components are less likely or able to do so. Specifically, fundamental principals limit the physical size of passive components such as transformers, resistors, capacitors and inductors, and while there is less need for numerous connectors in modern equipment, those that remain have physical restrictions on how small they can be and where they must be placed on a board. This has its benefits, however, as there now exist numerous 3D models for standard components such as passives and connectors that can be used and manipulated in a growing number of CAD packages.

The widespread creation of these 3D models indicates renewed commitment from vendors towards E-CAD/M-CAD integration; a trend that many in the industry believe will continue, bringing significant productivity gains to engineers across both domains."

—Frank Krämer, Technical Marketing Director, EMEA Altium Europe GmbH

In such cases, mechanical design dictates both the domains, yet there remains limited interaction between CAD tools targeting the mechanical and electronic domains."

Altium has taken the above to

heart and taken giant strides to add those features to Altium Designer's growing list of capabilities. ●

The author is a tech correspondent at EFY Bengaluru

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